Correspondence between transfers in a cap-and-trade and differentiated carbon prices

Contents

1 Introduction

The Paris Agreement commits its signatories to limiting global warming to 1.5–2°C above pre-industrial levels. Meeting that target requires cutting global CO_2 emissions by roughly 28 $GtCO_2$ per year relative to current policy trajectories, yet the 2025 round of Nationally Determined Contributions (NDCs) closes barely fourteen percent of that gap {unepgap2024}. It can be argued that this shortfall is a failure of coordination with a distributional disagreement that has never been cleanly resolved.

The disagreement takes the following form. Lower-income economies, whose per-capita emissions are below the global average, argue that a high, uniform global carbon price would impose disproportionate adjustment costs on their populations and their still-developing industrial bases. They therefore claim the right to set lower, nationally differentiated prices, or to delay ambitious action until wealthier countries provide sufficient financial compensation. Proponents of differentiated prices appeal to equity: a country that cannot afford to decarbonise its industrial base should not face the same carbon price as a wealthy one. However, critics of this view counter that differentiated prices are economically inefficient and that equity objectives are better achieved through lump-sum transfers alongside a uniform price.

This paper argues that the disagreement rests on a false dichotomy. The central contribution is a distributional equivalence result: for any system of nationally differentiated carbon prices in autarky—that is, without international transfers—there exists a uniform global carbon price combined with appropriately calibrated emission rights that delivers the same welfare outcome for every country, to a first-order approximation. Because abatement costs are convex, the uniform-price regime is moreover strictly more efficient: international carbon trading always weakly dominates autarky. The debate over prices versus rights is therefore not a debate about fundamentally different policy instruments but about the distribution of a single policy's gains, and it can in principle be resolved by choosing the right allocation of emission rights.

This equivalence holds cleanly in a static, single-period setting. It breaks down once time is introduced, because the welfare value of an emission right then depends not only on its quantity but on the full trajectory of the future carbon price, a trajectory that cannot be recovered analytically without knowledge of how capital markets, abatement technologies, and emissions elasticities evolve. We characterise two special cases that restore a partial equivalence: first, the case in which country-level autarky prices are homothetic to the global price path and each country receives a fixed share of the global carbon budget; and second, the more restrictive but analytically tractable Hotelling case, in which efficient capital markets and perfect foresight keep discounted carbon prices constant over time. In the Hotelling case the dynamic formula collapses to one that is structurally identical to the static result, with the intertemporal carbon budget playing the role of the single-period emission right.

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This paper connects to several strands of the literature. The efficiency comparison between carbon prices and quantity instruments originates with {weitzman1974prices} and has been extended in many directions. Our contribution is different, in that we are not comparing instruments for their efficiency alone but establishing their distributional equivalence and then quantifying the efficiency cost of departing from it. The political economy of cap-and-trade versus carbon taxes is examined in {goulder2013carbon} and {stavins2019carbon}; again, our angle is complementary. A growing literature uses integrated assessment models to study distributional consequences of climate policy across countries {fabre2024uniform}. We build directly on the NICE model {youngbrun2025} and extend it by constructing the two-dimensional parameter space of autarky price factors and emission-rights multipliers over which the equivalence can be tested. The efficiency of uniform carbon pricing as a corollary of the Diamond–Mirrlees production efficiency theorem {diamond1971optimal} is well known in principle. We make the argument precise in the context of international climate policy and attach quantitative stakes to it.

The remainder of the paper is organised as follows. Section 2 derives the static distributional equivalence and the Diamond–Mirrlees efficiency result, then characterises the conditions under which a partial equivalence extends to the dynamic case. Section **??** describes the simulation design. Section **??** presents the heatmaps and assesses where the Hotelling approximation holds and where it fails.

2 Theory

This section establishes the core theoretical claims of the paper. We begin with a static, single-period model in which the equivalence between differentiated prices and uniform prices with transfers takes a closed, analytically tractable form. We then extend the analysis to a dynamic setting and show that the correspondence breaks down, because the welfare value of an emission right depends on the price trajectory rather than on quantities alone. We characterise two special cases that restore a partial equivalence, homothetic prices with a fixed budget share, and the Hotelling case, and explain why the Hotelling case serves as the natural empirical benchmark for the simulations that follow.

2.1 The static case

Setting. Consider a world of many countries indexed by i, a single period, and no uncertainty. Every country has an incentive to reduce its emissions, but decarbonisation is costly. When country i faces a carbon price p_i, firms substitute towards low-carbon capital and households shift to less preferred, less carbon-intensive consumption. We call these abatement costs and denote them a_i. Allowing for international transfers t_i (positive if received, negative if paid) and abstracting from exogenous investment, country i's per-capita consumption is:

$$c_i = y_i + t_i - a_i, \tag{1}$$

where y_i is potential output per capita.

Welfare of country i depends on its consumption c_i and on global emissions $E = \sum_j e_j n_j$, where e_j is country j's per-capita emissions and n_j its population. We fix global emissions at their optimal level E^*: the question we ask is not "what is the optimal target?" but "given an agreed global target, which policy regime distributes its costs and benefits most fairly and efficiently?" The temperature outcome is identical under both regimes, only distribution and efficiency differ.

Let p^* be the uniform carbon price that attains E^*, defined implicitly by $E^* = \sum_j e_j(p^*)n_j$. We use the shorthands $e_i := e_i(p_i)$, $a_i := a_i(e_i)$, $e_i^* := e_i(p^*)$, and $a_i^* := a_i(e_i^*)$. A situation for country i, given fixed global emissions, can be summarised as $s = (t_i, p_i)$, and we write welfare as $u_i(t_i, p_i)$ with a slight abuse of notation.

The equivalent variation from the benchmark. To analyse an arbitrary situation s, it is useful to anchor the analysis to two special cases. The first is the benchmark situation $s^* = (0, p^*)$: no transfers and a uniform carbon price. The second is the autarky situation $s^a = (0, p_i)$: no transfers and a nationally differentiated price. We define $V_i(p_i)$ as the equivalent variation between autarky and the benchmark. It is the lump-sum transfer that, if paid at the benchmark price p^*, would make country i exactly as well off as under autarky:

$$u_i\Big(V_i(p_i),\, p^*\Big) = u_i(0,\, p_i). \tag{2}$$

Since consumption under autarky is $c_i^a = y_i - a_i$ and under the benchmark is $c_i^* = y_i - a_i^*$, we have:

$$V_i(p_i) = c_i^a - c_i^* = a_i^* - a_i. \tag{3}$$

This difference in abatement costs can be approximated by a first-order Taylor expansion around the benchmark emissions level e_i^*. At e_i^*, the optimising country sets its marginal abatement cost equal to the price it faces. Under p^* this gives $\frac{da_i}{de}(e_i^*) = -p^*$. Therefore:

$$V_i(p_i) \approx (e_i - e_i^*)\, p^*. \tag{4}$$

Denoting by $\varepsilon^* = -\frac{de_i/e_i^*}{dp/p^*}$ the price elasticity of emissions evaluated at p^*, we can write the equivalent variation more explicitly as:

$$V_i(p_i) \approx (p^* - p_i)\, e_i^*\, \varepsilon^*. \tag{5}$$

The sign is intuitive: if $p_i < p^*$, country i abates less in autarky and therefore has lower abatement costs. Moving it to the benchmark forces higher abatement, so a positive compensation $V_i > 0$ is required to hold welfare constant. The magnitude of the compensation is larger when the price gap is large, when baseline emissions are high, and when emissions are price-elastic. All three make the cost of tightening the price more burdensome.

The equivalent transfer for an arbitrary situation. Let g_i^s denote country i's welfare gain in situation $s = (t_i, p_i)$ relative to the benchmark, expressed as a money-equivalent transfer at p^*. We call this the equivalent transfer. It decomposes cleanly as:

$$g_i^s = t_i + V_i(p_i). \tag{6}$$

The first term is the actual transfer received, the second is the welfare gain or loss from facing a price that differs from the benchmark. Because g_i^s reduces every situation to a single money-equivalent number at a common price, it is the natural metric for comparing welfare across regimes that differ both in transfers and in prices.[1]

[1] One can also define an equivalent price p_i^* such that $(0, p_i^*)$ is welfare-equivalent to (t_i, p_i). Solving $t_i + (p^* - p_i)e_i^*\varepsilon^* \approx (p^* - p_i^*)e_i^*\varepsilon^*$ gives $p_i^* \approx p_i - t_i/(e_i^*\varepsilon^*)$. This equivalent price can be negative when the transfer is large, however, which has no natural economic interpretation as a carbon price. The equivalent transfer g_i^s does not share this limitation, which is why we use it as our primary welfare metric.

Correspondence between differentiated prices and emission rights. We can now state the central result of the static case. Suppose countries establish a cap-and-trade at the uniform price p^*, and each person in country i is allocated emission rights r_i. The net monetary transfer to i under the cap-and-trade is:

$$t_i^* = (r_i - e_i^*)\, p^*. \tag{7}$$

Country i sells rights if $r_i > e_i^*$ (it was allocated more than it emits at p^*) and buys them otherwise. The equivalent transfer in the cap-and-trade situation is therefore $g_i^* = t_i^*$. The equivalent transfer in autarky is, from equation (5):

$$g_i^a = V_i(p_i) \approx (p^* - p_i)\, e_i^*\, \varepsilon^*. \tag{8}$$

Setting $g_i^* = g_i^a$ and solving yields the equivalence relation:

$$\boxed{(r_i - e_i^*)\, p^* \approx (p^* - p_i)\, e_i^*\, \varepsilon^*,} \tag{9}$$

from which one can isolate either variable:

$$r_i \approx \left(1 + \left(1 - \tfrac{p_i}{p^*}\right)\varepsilon^*\right) e_i^*, \tag{10}$$

$$p_i \approx \left(1 + \tfrac{e_i^* - r_i}{e_i^*\, \varepsilon^*}\right) p^*. \tag{11}$$

Equation (10) is the key formula for policy design: given a country's autarky price p_i, it tells us how many emission rights to allocate in a cap-and-trade in order to leave that country's welfare unchanged. A country with a low autarky price ($p_i < p^*$) needs a generous rights allocation ($r_i > e_i^*$) to compensate for the abatement burden it takes on when forced to the higher uniform price; a country with a high autarky price ($p_i > p^*$) can be left with fewer rights than its equilibrium emissions.

Efficiency of the uniform price. The equivalence result shows that the two regimes can deliver identical welfare distributions, and we now want to see if the cap-and-trade do strictly better.

Suppose we give country i rights equal to its autarky emissions e_i, so that the net transfer under the cap-and-trade is $\tilde{t}_i = (e_i - e_i^*)p^*$. Convexity of abatement costs implies:

$$a_i - a_i^* \geq (e_i - e_i^*)\frac{da_i^*}{de} = (e_i - e_i^*)(-p^*), \tag{12}$$

and therefore $\tilde{t}_i = (e_i - e_i^*)p^* \geq a_i^* - a_i = V_i(p_i) = g_i^a$. Country i's consumption under the cap-and-trade is:

$$\tilde{c}_i = y_i + \tilde{t}_i - a_i^* \geq y_i - a_i = c_i^a. \tag{13}$$

Welfare is therefore weakly higher under the cap-and-trade than in autarky, for every country simultaneously. In autarky, a lower price reduces abatement costs, but at a diminishing rate, because of convexity. Selling spare permits on the cap-and-trade market generates revenue at the full market price p^*, which is always at least as good a deal as facing a lower domestic price. Trading strictly dominates autarky whenever abatement costs are strictly convex and prices differ across countries.

This is a statement of the Diamond–Mirrlees production efficiency theorem [?] applied to international climate policy: in the absence of other distortions, direct transfers are always preferable to differentiated factor prices as a redistributive instrument.

2.2 The dynamic case

Setting. We now add a time index t running over multiple periods. Each variable becomes a trajectory: prices $(p_{it})_t$, transfers $(t_{it})_t$, emissions $(e_{it})_t$. Let β_t be the discount factor between the initial period and period t. Intertemporal quantities are written in upper case: $G_i = \sum_t g_{it}\beta_t$ is the total discounted equivalent gain, and $R_i = \sum_t r_{it}$ is country i's intertemporal carbon budget.

Welfare equivalence now requires equality of entire discounted streams rather than point-in-time values. The natural dynamic extension of the static formula (9) is:

$$G_i^U = G_i^{\mathcal{A}} \iff \sum_t (r_{it} - e_{it}^*)\, \beta_t\, p_t^* \approx \sum_t (p_t^* - p_{it})\, e_{it}^*\, \varepsilon_t^*\, \beta_t. \tag{14}$$

In the static case, equation (9) admits a clean solution because p^* appears as a common factor on both sides and cancels. In equation (14) it does not: the price p_t^* varies over time and sits inside both sums, so the welfare value of a unit of emission rights in period t depends on what the carbon price happens to be in that period. A generous allocation in a period of low prices is worth less than the same allocation in a period of high prices. The upshot is that there is no direct correspondence between a country's intertemporal carbon budget R_i and its welfare. The timing of the allocation matters, and the timing depends on the full price trajectory $(p_t^*)_t$.

Special case 1: homothetic prices and fixed budget shares. Suppose the autarky price in each country is always proportional to the global benchmark price: $p_{it} = \pi_i p_t^*$ for a country-specific scalar π_i. Suppose also that country i receives a fixed share of the global emission budget in every period: $r_{it} = \rho_i r_t$, where r_t is the aggregate global allocation. Under these two assumptions, equation (14) simplifies to:

$$\rho_i\, \mathscr{B} - \sum_t e_{it}^*\, p_t^*\, \beta_t \approx (1 - \pi_i) \sum_t e_{it}^*\, \varepsilon_t^*\, p_t^*\, \beta_t, \tag{15}$$

where $\mathscr{B} = \sum_t r_t\, \beta_t\, p_t^*$ is a price-weighted sum of the global allocation. The formula is analogous to the static case but contains $\mathscr{B}$, which requires knowledge of the full price trajectory $(p_t^*)_t$. The equivalence holds, but it is not analytically tractable without a model.

Special case 2: the Hotelling case. In a world with a fixed aggregate carbon budget, perfect foresight, and efficient capital markets, the carbon price rises at the rate of return on capital. When we discount at that same rate, the discounted price $\beta_t p_t^*$ is constant at p_0 in every period. This means p_t^* factors out of the discounted sums in equation (14), and the equivalence reduces to:

$$\boxed{R_i - E_i^* \approx (1 - \pi_i) \sum_t e_{it}^*\, \varepsilon_t^*,} \tag{16}$$

where $E_i^* = \sum_t e_{it}^*$ is country i's total discounted emissions under the benchmark price trajectory. This is structurally identical to the static formula (9): the intertemporal carbon budget R_i plays the role of the single-period rights allocation, and the homothetic autarky price factor π_i plays the role of p_i/p^*. The timing of the rights allocation no longer matters, because every period carries the same discounted price.

The Hotelling case is the best possible setting for the equivalence: it requires the most favourable conditions (perfect foresight, efficient capital markets, a fixed global budget, and homothetic autarky prices). If the approximation fails even here, it will fail everywhere. This is why Section ?? uses equation (16) as the empirical benchmark: the simulations ask not whether the Hotelling formula holds in theory, but how far the numerically computed welfare effects deviate from what the formula predicts, and what those deviations reveal about the underlying mechanisms.

3 Methodology

3.1 Framework and parameters

- Policy timeline: 2030–2100 + disaggregation at the country–decile–year level
- Baseline carbon-price path (p_t^*): calibrated to enforce 1.8 °C global carbon budget, reaching net-zero global emissions by 2080 and net-negative after
- Welfare metric: Net Present Value (NPV) of EDE consumption, discounted at $r = 3\%$ + inequality aversion parameter $\eta = 1.5$.

3.2 Scenario 1: Autarky baseline

- Country i sets an independent domestic carbon price $p_i = \pi_i \times p^*$
- International revenue pooling and transfers disabled (revenues recycled within the country)
- Rest of the World (RoW) price p_{-i} computed dynamically from the global price identity:
  $$p^* = \omega_i p_i + (1 - \omega_i) p_{-i}$$
  where ω_i is country i's dynamic share of global baseline emissions
- We make sure that the global emissions cap is maintained across the simulation (Newton and bisection)

3.3 Scenario 2: Uniform price with varying rights

- All regions face the uniform global price p_t^* but emission rights vary by country
- Country i receives a share of the total global carbon budget proportional to its population, scaled by the rights multiplier ρ_i
- The remaining global budget is distributed to the RoW proportionally based on baseline emissions (grandfathering) to prevent macroeconomic collapse of heavy emitters in the aggregate group

3.4 Building the heatmaps

- For each (ρ_i, π_i) combination, compute relative welfare variation:
  $$\Delta W_i = \frac{NPV_{\text{uniform}} - NPV_{\text{autarky}}}{NPV_{\text{autarky}}} \times 100$$

- Plot as a heatmap and trace indifference curve where the country experiences no net welfare loss

3.5 Flags

- Two conditions evaluated at every year of 2030–2100:
  1. $\omega_i \times \pi_i > 1$: this means that country i claims more than 100% of the global price burden, forcing $p_{-i} < 0$ (a market-distorting fossil-fuel subsidy for the RoW)
  2. $\rho_i > 1/\text{share of world pop} \iff \text{share of global rights} > 1$: granting country i its rights multiplier uses more than 100% of the total global carbon budget, forcing the RoW into negative emission rights
- If either condition is triggered in any single year, the parameter combination is flagged with a gray cross.

4 Results

An illustration of the numbers we're talking about:

Table 1: Autarky scenario for the USA ($\pi = 2$)

Variable	**2030**	**2040**	**2050**	**2060**	**2070**	**2080**	**2090**	**2100**
p^* (USD/tCO_2)	42.6	112.9	205.9	285.7	401.6	480.9	476.2	471.4
p_i (USD/tCO_2)	85.2	225.8	411.7	571.4	803.3	961.9	952.3	942.8
E_i (GtCO_2)	3.88	2.32	0.57	-0.48	-1.58	-2.01	-1.68	-1.37

Table 2: Uniform price scenario for the COD ($\rho = 5$)

Variable	**2030**	**2040**	**2050**	**2060**	**2070**	**2080**	**2090**	**2100**
p^* (USD/tCO_2)	42.6	112.9	205.9	285.7	401.6	480.9	476.2	471.4
r_i (GtCO_2)	2.211	1.854	1.313	0.989	0.427	0.0	0.0	0.0
E_i (GtCO_2)	0.005	0.008	0.009	0.009	0.005	0.0	0.0	0.0
Transfer (bn USD)	93.9	208.4	268.5	280.0	169.8	0.0	0.0	0.0

4.1 Reading the heatmaps

- x-axis: homothetic price factor π_i (autarky price relative to global price)
- y-axis: fixed rights-share multiplier ρ_i
- Colours: relative welfare variation ΔW_i (%)
- Solid line: empirical indifference curve
- Grey crosses: economical or mathematical extreme parameter combinations

- Dotted lines point to non-losing rights allocation (ρ_i that verifies $\pi_i = 1$)
- Grey crosses are drawn when $\omega_i \times \pi_i > 1$ (x-axis) or when $\rho_i > 1$/world pop. share (y-axis)
- These are parameter combinations for which no non-negative price p_{-i} satisfies the global price identity (and the static efficiency result breaks down)
- Countries with a larger share of world emissions or world population have more constrained feasible regions
- Example:
  - China and India (share of world pop. $> 10\%$): grey crosses appear from $\rho_i = 10$ on the y-axis